
NanoWasm Specification

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CONTENTS

1	Abstract Syntax	3
2	Validation	5
2.1	nop	5
2.2	drop	5
2.3	select	5
2.4	const	5
2.5	local.get	5
2.6	local.set	6
2.7	global.get	6
2.8	global.set	6
3	Execution	7
3.1	nop	7
3.2	drop	7
3.3	select	7
3.4	local.get <i>x</i>	8
3.5	local.set <i>x</i>	8
3.6	global.get <i>x</i>	8
3.7	global.set <i>x</i>	8
4	Binary Format	9

NanoWasm is a small language with simple types and instructions.

ABSTRACT SYNTAX

The *abstract syntax* of types is as follows:

$$\begin{aligned} \text{mut} &::= \text{mut} \\ \text{valtype} &::= \text{i32} \mid \text{i64} \mid \text{f32} \mid \text{f64} \\ \text{functype} &::= \text{valtype}^* \rightarrow \text{valtype}^* \\ \text{globaltype} &::= \text{mut}^? \text{valtype} \end{aligned}$$

Instructions take the following form:

$$\begin{aligned} \text{const} &::= 0 \mid 1 \mid 2 \mid \dots \\ \text{instr} &::= \begin{array}{l} \text{nop} \\ | \\ \text{drop} \\ | \\ \text{select} \\ | \\ \text{valtype.const } \text{const} \\ | \\ \text{local.get } \text{localidx} \\ | \\ \text{local.set } \text{localidx} \\ | \\ \text{global.get } \text{globalidx} \\ | \\ \text{global.set } \text{globalidx} \end{array} \end{aligned}$$

The instruction `nop` does nothing, `drop` removes an operand from the stack, `select` picks one of two operands depending on a condition value. The instruction `t.const c` pushed the constant `c` to the stack. The remaining instructions access local and global variables.

VALIDATION

NanoWasm instructions are *type-checked* under a context that assigns types to indices:

$$\text{context} ::= \{\text{globals } \text{globaltype}^*, \text{locals } \text{valtype}^*\}$$

2.1 nop

`nop` is **valid** with $\epsilon \rightarrow \epsilon$.

$$\overline{C \vdash \text{nop} : \epsilon \rightarrow \epsilon}$$

2.2 drop

`drop` is **valid** with $t \rightarrow \epsilon$.

$$\overline{C \vdash \text{drop} : t \rightarrow \epsilon}$$

2.3 select

`select` is **valid** with $t \ t \ \text{i32} \rightarrow t$.

$$\overline{C \vdash \text{select} : t \ t \ \text{i32} \rightarrow t}$$

2.4 const

`(t.const c)` is **valid** with $\epsilon \rightarrow t$.

$$\overline{C \vdash t.\text{const } c : \epsilon \rightarrow t}$$

2.5 local.get

`(local.get x)` is **valid** with $\epsilon \rightarrow t$ if:

- $C.\text{locals}[x]$ exists.
- $C.\text{locals}[x]$ is of the form t .

$$\frac{C.\text{locals}[x] = t}{C \vdash \text{local.get } x : \epsilon \rightarrow t}$$

2.6 local.set

(local.set x) is **valid** with $t \rightarrow \epsilon$ if:

- $C.\text{locals}[x]$ exists.
- $C.\text{locals}[x]$ is of the form t .

$$\frac{C.\text{locals}[x] = t}{C \vdash \text{local.set } x : t \rightarrow \epsilon}$$

2.7 global.get

(global.get x) is **valid** with $\epsilon \rightarrow t$ if:

- $C.\text{globals}[x]$ exists.
- $C.\text{globals}[x]$ is of the form $(\text{mut}^? t)$.

$$\frac{C.\text{globals}[x] = \text{mut}^? t}{C \vdash \text{global.get } x : \epsilon \rightarrow t}$$

2.8 global.set

(global.set x) is **valid** with $t \rightarrow \epsilon$ if:

- $C.\text{globals}[x]$ exists.
- $C.\text{globals}[x]$ is of the form $(\text{mut } t)$.

$$\frac{C.\text{globals}[x] = \text{mut } t}{C \vdash \text{global.set } x : t \rightarrow \epsilon}$$

EXECUTION

NanoWasm execution requires a suitable definition of state and configuration:

$$\begin{aligned} \text{addr} &::= 0 \mid 1 \mid 2 \mid \dots \\ \text{moduleinst} &::= \{\text{globals } \text{addr}^*\} \\ \text{val} &::= \text{const } \text{valtype } \text{const} \\ \text{store} &::= \{\text{globals } \text{val}^*\} \\ \text{frame} &::= \{\text{locals } \text{val}^*, \text{module } \text{moduleinst}\} \\ \text{state} &::= \text{store}; \text{frame} \\ \text{config} &::= \text{state}; \text{instr}^* \end{aligned}$$

We define the following auxiliary functions for accessing and updating the state:

$$\begin{aligned} \text{local}((s; f), x) &= f.\text{locals}[x] \\ \text{global}((s; f), x) &= s.\text{globals}[f.\text{module.globals}[x]] \\ \text{update}_{\text{local}}((s; f), x, v) &= s; f[.\text{locals}[x] = v] \\ \text{update}_{\text{global}}((s; f), x, v) &= s[\text{globals}[f.\text{module.globals}[x]] = v]; f \end{aligned}$$

With that, execution is defined as follows:

3.1 nop

1. Do nothing.

$$\text{nop} \hookrightarrow \epsilon$$

3.2 drop

1. Assert: Due to validation, a value is on the top of the stack.
2. Pop the value val from the stack.

$$\text{val drop} \hookrightarrow \epsilon$$

3.3 select

1. Assert: Due to validation, a value of valtype i32 is on the top of the stack.
2. Pop the value (i32.const c) from the stack.
3. Assert: Due to validation, a value is on the top of the stack.
4. Pop the value val_2 from the stack.
5. Assert: Due to validation, a value is on the top of the stack.
6. Pop the value val_1 from the stack.

7. If $c \neq 0$, then:

a. Push the value val_1 to the stack.

8. Else:

a. Push the value val_2 to the stack.

$$val_1 \ val_2 \ (i32.const \ c) \ select \ \hookrightarrow \ val_1 \ \text{if } c \neq 0$$

$$val_1 \ val_2 \ (i32.const \ c) \ select \ \hookrightarrow \ val_2 \ \text{otherwise}$$

3.4 local.get x

1. Let z be the current state.

2. Let val be $local(z, x)$.

3. Push the value val to the stack.

$$z; (local.get \ x) \ \hookrightarrow \ z; val \ \text{if } val = local(z, x)$$

3.5 local.set x

1. Assert: Due to validation, a value is on the top of the stack.

2. Pop the value val from the stack.

$$z; val \ (local.set \ x) \ \hookrightarrow \ z'; \epsilon \ \text{if } z' = update_{local}(z, x, val)$$

3.6 global.get x

1. Let z be the current state.

2. Let val be $global(z, x)$.

3. Push the value val to the stack.

$$z; (global.get \ x) \ \hookrightarrow \ z; val \ \text{if } val = global(z, x)$$

3.7 global.set x

1. Assert: Due to validation, a value is on the top of the stack.

2. Pop the value val from the stack.

$$z; val \ (global.set \ x) \ \hookrightarrow \ z'; \epsilon \ \text{if } z' = update_{global}(z, x, val)$$

BINARY FORMAT

The following grammars define the binary representation of NanoWasm programs.

First, constants are represented in LEB format:

$$\begin{aligned}
 \text{byte} &::= 0x00 \mid \dots \mid 0xFF \\
 u(N) &::= n:\text{byte} \Rightarrow n & \text{if } n < 2^7 \wedge n < 2^N \\
 &\mid n:\text{byte } m:u(N-7) \Rightarrow 2^7 \cdot m + (n - 2^7) & \text{if } n \geq 2^7 \wedge N > 7 \\
 u32 &::= u(32) \\
 u64 &::= u(64) \\
 f(N) &::= b*:\text{byte}^{N/8} \Rightarrow \text{float}(N, b*) \\
 f32 &::= f(32) \\
 f64 &::= f(64)
 \end{aligned}$$

Types are encoded as follows:

$$\begin{aligned}
 \text{valtype} &::= 0x7F & \Rightarrow i32 \\
 &\mid 0x7E & \Rightarrow i64 \\
 &\mid 0x7D & \Rightarrow f32 \\
 &\mid 0x7C & \Rightarrow f64 \\
 \text{mut} &::= 0x00 & \Rightarrow \epsilon \\
 &\mid 0x01 & \Rightarrow \text{mut} \\
 \text{globaltype} &::= t:\text{valtype } mut:\text{mut} & \Rightarrow mut \ t \\
 \text{resulttype} &::= n:u32 (t:\text{valtype})^n & \Rightarrow t^n \\
 \text{functype} &::= 0x60 \ t_1^*:\text{resulttype } t_2^*:\text{resulttype} & \Rightarrow t_1^* \rightarrow t_2^*
 \end{aligned}$$

Finally, instruction opcodes:

$$\begin{aligned}
 \text{globalidx} &::= x:u32 & \Rightarrow x \\
 \text{localidx} &::= x:u32 & \Rightarrow x \\
 \text{instr} &::= 0x01 & \Rightarrow \text{nop} \\
 &\mid 0x1A & \Rightarrow \text{drop} \\
 &\mid 0x1B & \Rightarrow \text{select} \\
 &\mid 0x20 \ x:\text{localidx} & \Rightarrow \text{local.get } x \\
 &\mid 0x21 \ x:\text{localidx} & \Rightarrow \text{local.set } x \\
 &\mid 0x23 \ x:\text{globalidx} & \Rightarrow \text{global.get } x \\
 &\mid 0x24 \ x:\text{globalidx} & \Rightarrow \text{global.set } x \\
 &\mid 0x41 \ n:u32 & \Rightarrow i32.\text{const } n \\
 &\mid 0x42 \ n:u64 & \Rightarrow i64.\text{const } n \\
 &\mid 0x43 \ p:f32 & \Rightarrow f32.\text{const } p \\
 &\mid 0x44 \ p:f64 & \Rightarrow f64.\text{const } p
 \end{aligned}$$